

1 PRODUCT OVERVIEW

outlierX360 is an industry-agnostic, ML-powered platform that continuously monitors process variables to proactively detect outliers, subtle process drifts, and potential leaks before conventional alarms trigger. It executes continuous fault detection on your existing plant data, autonomously classifying deviations as instrument drift, genuine faults, or expected baseline shifts to completely eliminate alarm fatigue. By delivering vital severity context for every event, it empowers engineering teams across any sector with a prioritized, evidence-based maintenance schedule.

01 Data and Configuration

Shown: the configure panel, then per-tag setup.

UPLOAD

FILTERS

CRITICAL TAGS

DISTURBANCES

RUN

DATA FILE

Olefin_cgc_2024.xlsx

DURATION

Last 90 days

PLANT - SUBSYSTEM

Olefin - CGC

GLOBAL DIRECTION

Both

Up

Down

TAG	DIRECTION	MIN	MAX	CRITICAL	DISTURBANCE
REACTOR_TEMP	Both	20.0	180.0	✓	—
CGC_SUCT_PRESS	Down	0.5	6.0	✓	—
AI-201_C2_BTMS	Up	0.0	2.5	✓	—
HX_OUT_TEMP	Both	380	460	✓	w = 0.28

Registry — plant and subsystem, results kept per pair

Tag controls — direction up/down/both, min-max filters, critical tags

Upload — wide-format xlsx or csv, timestamp plus numeric tag columns

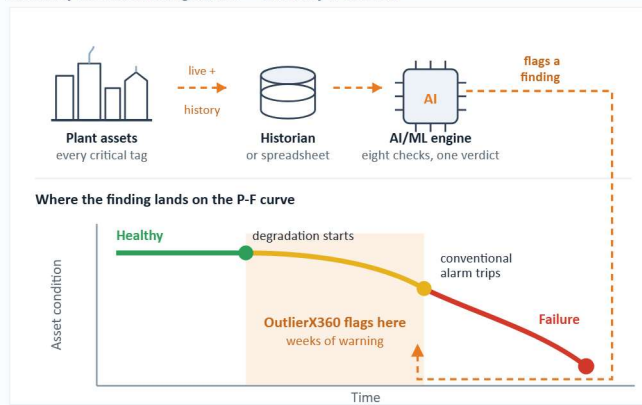
Disturbances — up to four known drivers, correlation-weighted

02 Detection and Diagnosis

Shown: how a developing failure is flagged, a real plant case, and the eight checks behind every finding.

HOW IT WORKS

Anomaly detection using AI/ML — and why it matters

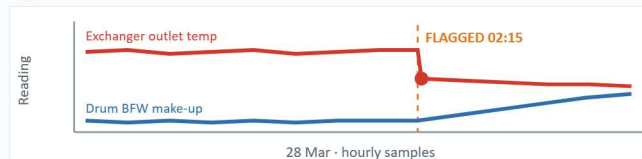


EARLY DETECTION WINDOW → TIME TO PLAN

REAL CASE

Heat Exchanger Tube Leak

Outlet temperature stepped down while drum BFW make-up climbed and AP held steady — a multi-tag leak signature, flagged on the first sample.



Flagged by: sudden jump · out of step with recent operation · out of line with related tags

EQUIPMENT FAULT → INSPECT EXCHANGER

THE EIGHT CHECKS

Detection Engine Suite

Eight independent checks on every tag; a finding is raised only when several agree.



S1 · Off normal level
outside the level this tag normally runs at



S2 · Out of step
no longer fits how the unit has been running



S3 · Extreme reading
well beyond the usual spread



S4 · Sudden jump
abrupt move between two readings



S5 · Out of line with peers
moves while related tags do not



S6 · Slow drift
gradual creep over days or weeks



S7 · New sustained level
settled apart from its own history



S8 · Drift since start-up
walk away from as-new performance

WHAT THE TEAM RECEIVES

Daily findings list

TAG	TIMESTAMP	CHECKS	READING	DIAGNOSIS
AI-201_C2_BTMS	14 Mar 06:40	3 of 8	1.84 mol%	Process issue
AI-201_PURITY	22 Mar 11:05	3 of 8	97.4 %	Tag issue
HX_OUT_TEMP	28 Mar 02:15	3 of 8	451 °C	Equipment fault
CGC_SUCT_PRESS	29 Mar 09:30	2 of 8	3.1 bar	Disturbance

Reason codes, correlated tags and expected values — exportable, delivered daily.

03 One Engine, Many Problems

One detection engine, many problem types — across process industries.

ONE ENGINE, SIX PROBLEM TYPES



APPLIED ACROSS INDUSTRIES

Petrochemicals	Refining	Power and utilities	Metals and mining
Specialty chemicals	Pharma and life sciences	Fertilizers, agrochem	Pulp, paper, cement

Any plant with time-stamped tag data — the same checks and diagnosis logic transfer unit to unit.

04 Business Impact

What early, diagnosed findings are worth to the operation.

Zero Loss of production Early alerts enable preventive action.	10x Faster fault identification Equipment faults split from sensor drift.
24x7 Always-on detection Every critical variable, around the clock.	100% Proactive posture Evidence-based maintenance planning.
<2% False positives Ends alarm fatigue, isolating genuine faults.	80% Less time chasing data One ranked list replaces trend hunting.

05 Platform

Runs inside your network; no data leaves the plant boundary.

Historian connectivity Plug and play with leading plant historians.	Live or on-demand Monitor continuously, or run a one-off review from a spreadsheet.
Daily reporting Downloadable daily reports with interactive trends.	Gen AI assistant Answers plant questions in the operating workflow.
Build your own dashboard Engineers build the views their unit needs.	Quick setup Configured and running within two weeks.

DELIVERY

Configured, validated and commissioned by Ingenero Data Science and Domain SMEs.

FLEXIBILITY

Any tag, any plant, any domain — across any industry.

SCALE

Plug in and use at any site, first-time or repeat — virtually no manual configuration required.

Contact for a free trial: marketing@ingenero.com